MPAS-URBAN QUICKSTART | GUIDE 2 OF 3

Initialize the HRAIN2025 case

Use the public 30 km–500 m grid or optionally regenerate it, then create static and initial-condition files.

OUTCOME You will have the static, initial-condition, and 112-task decomposition files needed to run the HRAIN2025 case.

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Before you begin

Complete Guide 1 and the recommended official MPAS tutorial first. Keep init_atmosphere_model, the standard MPAS static data, and the CGLC-LCZ dataset available. BNU is optional: retain the default soil categories supported by the selected source unless you deliberately choose the publicly downloadable BNU data. A fully comparable HRAIN2025 simulation requires the Dy and Fung (2016) soil dataset used for that case.

CASE DESCRIPTION - INTERCOMPARISON GUIDELINES V1.1
Event: The 2025 Black Rainstorm Event - HRAIN2025
Timespan: August 2-5, 2025, with the peak on August 5
Observation highlights: 358.8 mm on August 5 (355.7 mm at HKO headquarters), the highest single-day August rainfall since 1884; fourth black rainstorm warning in eight days; more than 9,600 cloud-to-ground lightning strikes between 05:00 and 12:00; peak rainfall near 90 mm h-1
Reported impacts: at least 2 deaths and more than 140 injuries; 42 elevator entrapments, 175 fire-alarm activations, 36 fallen trees, 7 landslides and 69 flooding cases; about 20% of flights cancelled and more than 400 delayed
Common simulation period: 2025-08-03 00:00 UTC to 2025-08-06 00:00 UTC (3 days)
GFS initialization used in this guide: 2025-08-03_00:00:00

For a first run, use the public grid and GFS intermediate file. Regenerating the global variable-resolution mesh is optional and requires JIGSAW, METIS, mpas_tools, GeoPandas, NumPy, xarray, SciPy, Shapely, and a high-memory compute node.

Step 1 | Download the ready grid and GFS input

RECOMMENDED BEGINNER PATH

```
export MPAS_ROOT="$HOME/MPAS"
mkdir -p "$MPAS_ROOT/HRAIN2025_INPUT"
cd "$MPAS_ROOT/HRAIN2025_INPUT"

curl -fL -O https://github.com/Liuzh223/MPAS-Urban-HRAIN2025/releases/download/v1.0/MPAS-Urban_HRAIN2025_30km-to-500m_grid-and-GFS2025080300_v1.0.tar.gz

echo "45fcfab80a3ecb5f8db558a4582d8910d1cac236f202ab499bb2b5e8e0e8143d  MPAS-Urban_HRAIN2025_30km-to-500m_grid-and-GFS2025080300_v1.0.tar.gz"   | sha256sum -c -

tar -xzf   MPAS-Urban_HRAIN2025_30km-to-500m_grid-and-GFS2025080300_v1.0.tar.gz

ls -lh 30km_500m.grid.nc GFS:2025-08-03_00
```

The archive contains only 30km_500m.grid.nc and GFS:2025-08-03_00. The GFS file is already in WPS intermediate format; do not run ungrib.exe on it again.

Step 2 | Optional: regenerate the Hong Kong mesh

Skip this step when using the public grid. Use it only when you want to change the refinement region or independently reproduce the mesh-generation workflow.

CHECK THE MESH ENVIRONMENT

```
command -v python
command -v gpmetis
python -c "import geopandas, numpy, xarray, scipy, shapely, mpas_tools"
```

GET THE CASE SOURCE

```
export MPAS_ROOT="$HOME/MPAS"
cd "$MPAS_ROOT"

# Skip this command if MPAS-Urban-HRAIN2025 already exists.
git clone https://github.com/Liuzh223/MPAS-Urban-HRAIN2025.git
cd "$MPAS_ROOT/MPAS-Urban-HRAIN2025"

curl -fL   https://raw.githubusercontent.com/pedrospeixoto/MPAS-BR/master/grids/utilities/jigsaw/jigsaw_util.py   -o mesh/jigsaw_util.py
```

Download the Hong Kong boundary through the Aliyun DataV area selector and save the exact file as mesh/hongkong.geojson: https://datav.aliyun.com/portal/school/atlas/area_selector . This is the source used by the case author. Aliyun DataV is a third-party service and is not an official Hong Kong SAR Government boundary source. Archive the exact GeoJSON for reproducibility, check its terms before redistribution, and note that another boundary can change the generated mesh.

GENERATE AND PARTITION

```
python mesh/generate_hk_500m_mesh.py   --geojson mesh/hongkong.geojson   --jigsaw-util mesh/jigsaw_util.py   --output-dir generated_mesh   --partitions 112

ls -lh generated_mesh/hk_hull_500m.mpas.nc
ls -lh generated_mesh/hk_hull_500m_graph.info.part.112
```

The generated hk_hull_500m.mpas.nc is an MPAS mesh file. The extension is only a filename convention. To use the same filename as the ready case:

```
cp generated_mesh/hk_hull_500m.mpas.nc    generated_mesh/30km_500m.grid.nc
```

MEMORY NOTE The default global sampling field is memory intensive. A failure caused by insufficient memory is not evidence that the ready Release grid is invalid; use the ready grid for the beginner workflow.

Step 3 | Create the case directory

Keep the compiled sources, datasets, repository, Release input, and independent case under one $HOME/MPAS root. MPAS_BUILD is the tutorial-defined shell variable introduced in Guide 1; it is not an official MPAS variable. Here it identifies the one compiled source tree that supplies the initialization executable and runtime files, while CASE remains separate from both source trees.

CHOOSE ONE COMPILED BUILD

```
export MPAS_ROOT="$HOME/MPAS"
export MPAS_BUILD="$MPAS_ROOT/HKUST-MPAS"

test -x "$MPAS_BUILD/init_atmosphere_model"
```

The standard HKUST-MPAS build is the default. For the verified HRAIN2025 case-specific build, replace MPAS_BUILD with $MPAS_ROOT/HKUST-MPAS-NTDK. Do not set both. The pinned cutoff build is commit 7c1610b8f41781c2433f780bd7496da63b25a920 and remains based on MPAS v8.2.2. This pinned source adds the config_soilcat_data option from MPAS v8.3+, allowing the optional publicly downloadable BNU data to be selected. If BNU is not selected, omit this option and retain the default soil categories supported by the source. Selecting BNU does not recreate the Dy and Fung (2016) dataset used for HRAIN2025.

CREATE AND ENTER THE INDEPENDENT CASE

```
export HRAIN_REPO="$MPAS_ROOT/MPAS-Urban-HRAIN2025"
export HRAIN_INPUT="$MPAS_ROOT/HRAIN2025_INPUT"
export CASE="$MPAS_ROOT/HRAIN2025_30km_500m"

mkdir -p "$CASE"
cd "$CASE"
```

COPY THE FILES THAT WILL BE EDITED

```
cp "$HRAIN_REPO/config/namelist.init_atmosphere" ./namelist.init_atmosphere
cp "$MPAS_BUILD/streams.init_atmosphere" ./streams.init_atmosphere
```

LINK THE SELECTED BUILD AND LARGE INPUTS

```
ln -sf "$MPAS_BUILD/init_atmosphere_model" ./init_atmosphere_model
ln -sf "$MPAS_BUILD/vertical_levels" ./vertical_levels
ln -sf "$HRAIN_INPUT/30km_500m.grid.nc" ./30km_500m.grid.nc
ln -sf "$HRAIN_INPUT/GFS:2025-08-03_00" ./GFS:2025-08-03_00
```

COPY THE PARTITION AND CHECK

```
cp "$HRAIN_REPO/mesh/hk_hull_500m_graph.info.part.112" ./hk_hull_500m_graph.info.part.112

ls -lh init_atmosphere_model
ls -lh namelist.init_atmosphere streams.init_atmosphere
ls -lh 30km_500m.grid.nc GFS:2025-08-03_00
ls -lh hk_hull_500m_graph.info.part.112
ls -lh vertical_levels/urban_ZR_75.txt
```

PATH NOTE The Release archive does not contain vertical_levels/urban_ZR_75.txt. It must exist in the selected MPAS_BUILD. Keep every linked runtime file from that same build. After outputs are created, do not relink this case to another build; create a separate case directory for a version comparison.

Optional soil-category selection

Dy and Fung (2016) soil dataset

HRAIN2025 used the updated global WRF soil map described by Dy and Fung (2016). The dataset is not included here. The paper data link resolves to a Google Drive folder; request access and processing details from the authors if necessary.

EXACT-COMPARISON NOTE The optional BNU setting does not recreate the Dy and Fung (2016) dataset. A fully comparable HRAIN2025 simulation requires the same soil dataset and initialization setup; consult the paper and contact its authors for the data and processing details.

Optional BNU soil data | Official MPAS download

BNU is optional. MPAS v8.3+ provides config_soilcat_data for selecting the official 30-arc-second BNU top categories. The pinned HRAIN2025 source remains based on MPAS v8.2.2 but adds this option from v8.3+. If you choose BNU, download and unpack the data as shown in Guide 1 and verify the index before starting the static stage. The BNU data are publicly downloadable and are similar to, but not identical to, the Dy and Fung (2016) dataset.

OPTIONAL BNU CHECK

```
test -s "$MPAS_ROOT/DATA/mpas_static/bnu_soiltype_top/index"
```

Step 4 | Create the reusable static file

Create a separate static-stage copy before editing. The HRAIN2025 reference namelist below is configured for meteorological initialization, so do not overwrite the only copy.

REFERENCE NAMELIST https://github.com/Liuzh223/MPAS-Urban-HRAIN2025/blob/main/config/namelist.init_atmosphere

FILE TO EDIT Copy namelist.init_atmosphere to namelist.init.static, then edit namelist.init.static for this step.

CHANGE 1 - &data_sources/config_geog_data_path: replace the example relative path with the real absolute directory containing the standard MPAS data and mpas_cglc_lcz.

CHANGE 2 - &data_sources/config_landuse_data: select 'CGLC_LCZ' for the urban static interpolation.

CHANGE 3 - &data_sources/config_soilcat_data (optional): use this option only when selecting BNU with MPAS v8.3+ or the pinned v8.2.2-based source, which adds config_soilcat_data from v8.3+. Standard hkust-dev v8.2.2 does not provide config_soilcat_data and can only use its default soil-category data

(STATSGO), so omit or comment out this line. The BNU data are publicly downloadable but are not identical to the Dy and Fung (2016) dataset used for HRAIN2025.

CHANGE 4 - &preproc_stages: use the official tutorial static-stage block below. Enable static and native-GWD processing, disable vertical-grid and meteorological interpolation, keep config_input_sst = false, and set config_frac_seaice = false.

ESSENTIAL SETTINGS

```
&nhyd_model
    config_init_case = 7
/

&data_sources
    config_geog_data_path = '/home/USER/MPAS/DATA/mpas_static/'
    config_landuse_data   = 'CGLC_LCZ'
    ! config_soilcat_data = 'BNU'   ! uncomment only for optional BNU
/

&preproc_stages
    config_static_interp     = true
    config_native_gwd_static = true
    config_vertical_grid     = false
    config_met_interp        = false
    config_input_sst         = false
    config_frac_seaice       = false
/
```

OPTIONAL BNU INPUT CHECK Run this only when BNU is selected with MPAS v8.3+ or the pinned v8.2.2-based source, which adds config_soilcat_data from v8.3+. It verifies the downloaded BNU files, not the Dy and Fung (2016) dataset.

```
test -s "$MPAS_ROOT/DATA/mpas_static/bnu_soiltype_top/index"
```

FILE TO EDIT streams.init_atmosphere. Set the input stream to the grid and the output stream to a new static file:

```
<streams>
<immutable_stream name="input"
                  type="input"
                  filename_template="30km_500m.grid.nc"
                  input_interval="initial_only" />
<immutable_stream name="output"
                  type="output"
                  filename_template="30km_500m.static.nc"
                  packages="initial_conds"
                  output_interval="initial_only" />
</streams>
```

EXPECTED WAIT - STATIC STAGE

Static-field generation is the slowest step. For this large global variable-resolution mesh, plan for several hours; 6-12 hours is a reasonable estimate for the tested single-process setup. Slower storage or limited memory may take longer. This is a planning estimate, not a guarantee.

RUN

```
cd "$CASE"
mpiexec -n 1 ./init_atmosphere_model

grep -E "Using (BNU|STATSGO).*soil type dataset" \
  log.init_atmosphere.0000.out
tail -n 40 log.init_atmosphere.0000.out
ls -lh *.static.nc
```

WHILE THE JOB IS RUNNING

Use a second terminal to monitor the log or plot the complete 30km_500m.grid.nc. Do not modify the run directory, start another init_atmosphere_model there, or read partial .static.nc or .init.nc files. A fully comparable HRAIN2025 simulation requires the Dy and Fung (2016) dataset; the publicly downloadable BNU data are similar but not identical. Never reuse a static file after changing the soil input.

```
tail -f log.init_atmosphere.0000.out
```

Step 5 | Create the initial condition

Restore a fresh copy of the linked HRAIN2025 namelist for meteorological initialization. Edit the fields below before running; do not reuse the static-stage flags.

REFERENCE NAMELIST https://github.com/Liuzh223/MPAS-Urban-HRAIN2025/blob/main/config/namelist.init_atmosphere

CHANGE 1 - &nhyd_model: config_start_time and config_stop_time must both be 2025-08-03_00:00:00.

CHANGE 2 - &data_sources: config_met_prefix = 'GFS' must match GFS:2025-08-03_00.

CHANGE 3 - &data_sources/config_soilcat_data: keep this option only if BNU was selected when creating the static file. Standard hkust-dev v8.2.2 does not provide config_soilcat_data and can only use its default soil-category data (STATSGO), so remove or comment out this line. This step does not change the soil categories already stored in the static file.

CHANGE 4 - &vertical_grid: config_specified_zeta_levels must point to an existing vertical_levels/urban_ZR_75.txt.

CHANGE 5 - &preproc_stages: use the complete meteorological-initialization block below. Switch off static and native-GWD generation, switch on vertical-grid and GFS interpolation, and keep config_input_sst = false and config_frac_seaice = true.

CHANGE 6 - &decomposition: keep the prefix hk_hull_500m_graph.info.part. so that 112 tasks select the supplied .part.112 file.

ESSENTIAL SETTINGS

```
&nhyd_model
    config_init_case  = 7
    config_start_time = '2025-08-03_00:00:00'
    config_stop_time  = '2025-08-03_00:00:00'
/

&data_sources
    config_met_prefix = 'GFS'
/

&preproc_stages
    config_static_interp     = false
    config_native_gwd_static = false
    config_vertical_grid     = true
    config_met_interp        = true
    config_input_sst         = false
    config_frac_seaice       = true
/

&decomposition
    config_block_decomp_file_prefix =
      'hk_hull_500m_graph.info.part.'
/
```

FILE TO EDIT streams.init_atmosphere. Use the static file as input and choose the initial-condition output:

```
<streams>
<immutable_stream name="input"
                  type="input"
                  filename_template="30km_500m.static.nc"
                  input_interval="initial_only" />
<immutable_stream name="output"
                  type="output"
                  filename_template="30km_500m.init.nc"
                  packages="initial_conds"
                  output_interval="initial_only" />
</streams>
```

EXPECTED WAIT - INITIAL-CONDITION STAGE

Allow about 10-30 minutes for this mesh; a busy cluster or slow shared filesystem may approach one hour. This is a planning estimate; runtime depends on the machine, MPI layout, memory, and filesystem.

RUN WITH THE MATCHING PARTITION

```
export OMP_NUM_THREADS=1
mpiexec -n 112 ./init_atmosphere_model
tail -n 50 log.init_atmosphere.0000.out
ls -lh *.init.nc
```

CHECK The MPI task count must match the .part.112 suffix. Do not use the single GFS file for another initialization time.

If you need another date

Follow the official MPAS/WPS tutorial to download the matching GFS GRIB2 files and run WPS ungrib.exe. Use the prefix GFS and provide every valid time required by the intended initialization. The HRAIN2025 Release file is only for 2025-08-03_00:00:00.

For WPS installation and ungrib.exe usage, see the official WRF Online Tutorial and the WRF Users' Guide - WPS. For this MPAS workflow, only the ungrib.exe procedure is relevant; geogrid.exe and metgrid.exe are not used.

This HRAIN2025 mesh is global with variable resolution, so the beginner workflow does not create limited-area lateral boundary files.

Ready for Guide 3

- The static file exists and is non-empty.
- The selected soil input is recorded. BNU is optional and is not identical to the Dy and Fung (2016) dataset used for HRAIN2025.
- The 2025-08-03_00:00:00 initial-condition file exists and is non-empty.
- The decomposition file ends in .part.112 and the planned run uses 112 MPI tasks.

Primary references

- Dy and Fung (2016), updated global WRF soil map
- Paper data folder (Google Drive; access may require sign-in)
- Official MPAS static datasets: https://www2.mmm.ucar.edu/projects/mpas/site/documentation/users_guide/appA.html
- Selectable BNU soil-category input: https://github.com/MPAS-Dev/MPAS-Model/pull/1322
- HRAIN2025: https://github.com/Liuzh223/MPAS-Urban-HRAIN2025
- Official MPAS tutorial: https://mpas-dev.github.io/atmosphere/tutorial.html
- Hong Kong boundary used by this case (third-party, not an official Hong Kong SAR Government source): https://datav.aliyun.com/portal/school/atlas/area_selector